Annexure III: Teaching Strategy

CLO No.	Teaching-Learning Activities for Active Learning	Assessment (Formative and Summative) Activities
CLO 1	1. Discussion 2. Problem Solving 3. Assignment on topics 4. Tutorial sessions	1. Continuous Assessment 2. Problem solving in the tutorial 3. Assignment and Case Study
CLO 2	1. Discussion 2. Reflection 3. Assignment on topics 4. Tutorial Sessions 5. Think Pair Share	1. Continuous Assessment 2. Problem solving in the tutorial 3. Assignment and Case Study
CLO 3	1. Discussion 2. Problem Solving 3. Assignment on topics 4. Tutorial Sessions	1. Continuous Assessment 2. Problem solving in tutorial 3. Assignment and Case study
CLO 4	1. Problem Solving 2. Assignment on topics 3. Tutorial Sessions	1. Continuous Assessment 2. Problem solving in tutorial 3. Assignment and Case study

Annexure IV: Course Outline Template

Institute	School of Technology, Institute of Technology
Department	Computer Science and Engineering
Programme	B.Tech.
Batch	AY: 2025-26 (Students Entry year: 2024-25)
Term/Semester	III
Course Title	Mathematical Foundations for Computer Science
Course Code	2CS802CC25/2CS802CC23/2CS802
Credit Hours	L-2, T-1, P-0, C-3
Faculty's Name	**Course Coordinator:** Prof Chandan Trivedi (CT) **(Div-C)** **Faculties:** Prof Bela Shrimali **(Div-A),** Prof Shanvendra Rai (NFSR) **(Div-B)**, Prof Deepa R (NFDR) **(Div-D),** Prof Jigna Patel (JAP) **(Div-E),** Prof Snehal Rajput (NFRS) **(Div-F),** Prof Lata Gohil (LG) **(Div-G),** Prof Aparna Kumari (AK) **(Div-D2D)**
E-mail ID	chandan.trivedi@nirmauni.ac.in, bela.shrimali@nirmauni.ac.in, shanvendra.rai@nirmauni.ac.in deepa.r@nirmauni.ac.in, jignas.patel@nirmauni.ac.in, snehal.rajput@nirmauni.ac.in lata.gohil@nirmauni.ac.in aparna.kumari@nirmauni.ac.in,
Contact No.	9714786338(CT), 9925222052 (SB), 9898942993(JAP), 9739463988(AK), 9426359352 (LG), 9537316314 (NFRS), 8088326583 (NFDR), (NFSR)
Office Hours	7:30 pm to 6:00 pm
Moodle Details (as and when available)	https://lms.nirmauni.ac.in/course/view.php?id=1067

A. **Course Overview** (300 to 1000 words)

Importance of the course

This course is intended to provide a refresher to students on several concepts of discrete mathematics and to understand their significance in various engineering applications.

Objective of the Course

The objective of the course is to build the fundamentals from a mathematical perspective in students and prepare them for domain-specific or interdisciplinary applications of these aspects. Students are expected to learn the concepts of various facets of foundation in discrete mathematics and use them in solving problems in relevant fields.

Pre-requisite

Basic knowledge of school-level mathematical concepts

Syllabus

Unit	Contents	Teaching Hours (Total 30)
Unit-I	**Basic concepts of Discrete Mathematics:** Sets, Set Operations, Functions and their types, Sequences and Summations, Cardinality of Sets, Relations and their Properties, n-ary Relations and their applications, Closures of Relations, Equivalence Relations, Partial Orderings, Pigeon-hole principle, the principle of inclusion and exclusion	08
Unit-II	**Propositional Logic**: syntax, semantics, validity and satisfiability, basic connectives and truth tables, Applications of Propositional Logics, Propositional Equivalences, Predicates and Quantifiers, Nested Quantifiers, Rules of Inference, Normal forms (CNF, DNF)	07
Unit-III	**Proof Techniques**: proof methods and strategies, direct proof, indirect proof (proof by contraposition), proof by contradiction, principles of mathematical induction, strong induction, the well-ordering principle	04
Unit-IV	**Algebraic Structures**: Introductions to groups, Examples of groups, basic algebra in groups, the order of group elements, cyclic groups and subgroups, cyclic groups, introduction to rings and fields	03
Unit-V	**Recurrence Relations:** Introduction to recurrence relations, Applications of Recurrence Relations, Solving linear homogeneous and non-homogeneous recurrence relations	03
Unit-VI	**Graphs:** Graphs and Graph Models, Graph Terminology and Special Types of Graphs, Representing Graphs and Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Planar Graphs, Graph Coloring **Trees:** Introduction to Trees, Applications of Trees, Spanning Trees, Minimum Spanning Trees, and its applications.	05

B. Course Learning Outcomes (CLOs)

At the end of the course, the students shall be able to:

1. define preliminaries of discrete mathematics, concepts of sets, graphs, digraphs and trees (BL1)
2. identify properties of relations, functions, equivalence, partial order relations, and recurrence relations (BL2)
3. apply proof techniques, mathematical induction, and recurrence techniques (BL3)
4. examine logical propositions and algebraic structures (BL4)

C. Text Book (reflect other study/reference materials in session plan)

1. C. L. Liu, “Elements of Discrete Mathematics”, McGraw-Hill
2. Kenneth H. Rosen, “Discrete Mathematics and its Applications”, McGraw-Hill
3. Norman L. Biggs, “Discrete Mathematics”, Oxford Press
4. Tremblay, J.P. & Manohar, Discrete mathematical structures with applications to computer science, McGraw-Hill
5. Susanna S. Epp, Discrete Mathematics with Applications, Wadsworth Publishing

Self-study:
As per the syllabus notification, 10% of the syllabus is to be identified as the self-study component, and a related assessment policy is to be declared. The aim of this exercise is to impart lifelong learning skills among budding professionals.

D. Assessment Components & Schedule

SEE	CE	LPW
0.4	0.6	-

Bifurcation of Components in CE and SEE

Component (with No. of sub-components)	Weightage (Each part of the Component)	Schedule	Overall Weightage of the Component	CLO Number
Class Test	30% of CE **(30 marks)** (Quiz1, Quiz2, and Quiz3 each of 15 marks) – Best 2 out of 3 will be considered	To be announced	0.3	1,2,3,4
Sessional Exam	40% of CE **(40 marks)** (Exam will be of 50 marks and will be converted to 40)	Tentative: 1-3 Sep, 2025	0.4	1,2,3,4
Case Study (Innovative Assignment)	30% of CE **(30 marks)** (Two tasks of the Case Study)	On or before 15 Oct, 2025	0.3	1,2,3,4

CE component Overall			**0.6**	
SEE	100% **(100 marks)**	To be declared	**0.4**	1,2,3,4

E. **Session Plan**

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
1	Overview of the course, Discussion on Course Policy, Course Website and Blog, Importance of the course, Evaluation, Linkages of the course with other courses, and Professional relevance	**Text Book**	NA
		Reading (Reference)	Course Policy Document
		Pedagogy	**Constructive Approach** (Student cantered)
		Mention CLO Number/s (may be more than one)	NA
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Understand the importance, scope, and policy of the course
		URL	https://lms.nirmauni.ac.in/course/view.php?id=1067

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
2	Operations and laws of sets, Cartesian products, Set identities, Cardinality of sets	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Integrative Approach (**Kindles interest in Mathematics and Science, specifically)
		Mention CLO Number/s (may be more than one)	1
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Understand set theory along with its real-life applications
		URL	https://www.javatpoint.com/sets-introduction https://www.geeksforgeeks.org/set-theory/?ref=lbp

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
3	Set theory and Venn diagram with examples,	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Constructive Approach (**Learning by doing)
		Mention CLO Number/s (may be more than one)	1, 4
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Understand importance of metrices and summation with its real-life applications
		URL	https://www.geeksforgeeks.org/set-theory/?ref=lbp

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
4	Functions and their types, composition	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Inquiry-Based Approach (**Enhances problem-solving skills)
		Mention CLO Number/s (may be more than one)	1
		Session Learning Outcome (SLO	Understand and correlate the applicability of functions in real-life applications
		URL	https://www.geeksforgeeks.org/introduction-to-domain-and-range-relations-and-functions/?ref=lbp https://www.javatpoint.com/functions

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
5	Relation and their properties, Representation, n-ary relations and their applications	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw Hill
		Pedagogy	**Constructive Approach (**Builds on past knowledge)
		Mention CLO Number/s (may be more than one)	2
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Conceptualize and relate the application of computer science
		URL	https://www.geeksforgeeks.org/introduction-to-domain-and-range-relations-and-functions/?ref=lbp

			https://www.javatpoint.com/representation-of-relations

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
6	Closures of relations, equivalence relations	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Inquiry-Based Approach (**Enhances problem-solving skills)
		Mention CLO Number/s (may be more than one)	2
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Understand the relations and its applicability in problem solving
		URL	https://www.javatpoint.com/representation-of-relations https://www.javatpoint.com/closure-properties-of-relations

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
7	Solving Examples on Sets, Functions, and Relations, Sequences and Summations	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Inquiry-Based Approach (**Enhances problem-solving skills)
		Mention CLO Number/s (may be more than one)	1 and 2
		Session Learning Outcome (SLO) [may	Understand importance of solving mathematical problems and apply set theory concept

		vary from 1 to 3]	
		URL	

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
8	Principle of Inclusion and Exclusion and pigeonhole principle	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw Hill
		Pedagogy	**Constructive Approach (**Learning by doing)
		Mention CLO Number/s (may be more than one)	1 and 2
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Apply knowledge of inclusion and exclusion and solve aptitude problems in in applications
		URL	https://www.javatpoint.com/pigeonhole-principle https://www.geeksforgeeks.org/inclusion-exclusion-various-applications/

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
9	Partial Orderings, Hasse Diagrams, Lattices, and their types	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Integrative Approach (**Encourages application of acquired knowledge)
		Mention CLO Number/s (may be more than one)	1 and 3
		Session Learning Outcome (SLO) [may	Realize need and importance of poset and lattices

		vary from 1 to 3]	Solve the basic problems of hasse diagram and related concept
		URL	https://www.geeksforgeeks.org/discrete-mathematics-hasse-diagrams/

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
10	Propositional Logic: syntax, semantics, validity, and satisfiability	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Integrative Approach (**Encourages application of acquired knowledge)
		Mention CLO Number/s (may be more than one)	1 and 3
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Realize need and importance of propositional logic, semantics, validity and satisfiability Solve the basic problems of logic development
		URL	https://www.geeksforgeeks.org/engineering-mathematics-tutorials/?ref=ghm#logic https://www.javatpoint.com/propositions-and-compound-statements

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
11	Basic connectives and truth tables	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Constructive Approach (**Learning by doing)
		Mention CLO Number/s (may be more than one)	1 and 3

		Session Learning Outcome (SLO) [may vary from 1 to 3]	Implement Basic connectives and truth tables
		URL	https://www.javatpoint.com/basic-logical-operations https://www.geeksforgeeks.org/mathematical-logic-introduction-propositional-logic-set-2/

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
12	Concept of logical equivalence, Applications of logic Normal Forms (CNF, DNF)	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Integrative Approach (**Encourages application of acquired knowledge)
		Mention CLO Number/s (may be more than one)	1 and 3
		Session Learning Outcome (SLO)	Apply the concept of logical equivalence
		URL	https://www.geeksforgeeks.org/mathematical-logic-propositional-equivalences/ https://www.javatpoint.com/tautologies-and-contradiction

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
13-14	laws of logic, logical implication, rules of inference	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Inquiry-Based Approach (**Excites curiosity in the learners)
		Mention CLO Number/s	1 and 3

		(may be more than one)	
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Understand laws of logic, logical implication, and rules of inference
		URL	https://www.javatpoint.com/tautologies-and-contradiction https://www.javatpoint.com/conditional-and-biconditional-statements

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
15	Predicates and Quantifiers	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Inquiry-Based Approach (**Enhances problem-solving skills)
		Mention CLO Number/s (may be more than one)	1 and 3
		Session Learning Outcome (SLO)	Apply the use of predicates and quantifiers
		URL	https://www.geeksforgeeks.org/mathematic-logic-predicates-quantifiers/?ref=lbp https://www.javatpoint.com/predicate-logic

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
16	Nested quantifiers and problems in logic	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Inquiry-Based Approach (**Enhances problem-solving skills)

		Mention CLO Number/s (may be more than one)	1,2,3
		Session Learning Outcome (SLO) [may vary from 1-3]	Apply logic and its operations on mathematical problems
		URL	https://www.javatpoint.com/predicate-logic https://www.geeksforgeeks.org/mathematics-theorems-nested-quantifiers/?ref=lbp

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
17	Proof methods and strategies, direct proof, proof by contraposition	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Inquiry-Based Approach (**Learning by doing)
		Mention CLO Number/s (may be more than one)	1 and 4
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Understand the concept of proof technique
		URL	https://math.stackexchange.com/questions/262828/using-proof-by-contradiction-vs-proof-of-the-contrapositive https://math.libretexts.org/Courses/Monroe_Community_College/MTH_220_Discrete_Math/3%3A_Proof_Techniques/3.4%3A_Indirect_Proofs

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
18	Proof by contradiction	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw Hill

		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw Hill
		Pedagogy	**Constructive Approach (**Learning by doing**)**
		Mention CLO Number/s (may be more than one)	1 and 4
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Understand the concept proof by contraposition
		URL	https://www2.edc.org/makingmath/mathtools/contradiction/contradiction.asp https://www.math-cs.gordon.edu/courses/mat231/notes/proof-contradiction.pdf

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
19	Proof Techniques: Mathematical Induction, Strong mathematical induction, and the well-ordering principle	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Constructive Approach (**Learning by doing)
		Mention CLO Number/s (may be more than one)	1 and 4
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Apply Proof Techniques: Mathematical Induction Understand the concept of strong mathematical induction and well-ordering principle
		URL	https://www.javatpoint.com/mathematical-induction https://www.tutorialspoint.com/discrete_mathematics/discrete_mathematical_induction.htm

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
20	Recursive definition, Proof of sufficiency and necessity	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Constructive Approach (**Learning by doing)
		Mention CLO Number/s (may be more than one)	1 and 4
		Session Learning Outcome (SLO) [may vary from 1-]	Understand the concept of proof of necessity and sufficiency Understand the concept recursive definition
		URL	https://mfleck.cs.illinois.edu/building-blocks/version-1.0/recursive-definition.pdf https://math.stackexchange.com/questions/4351831/necessary-or-sufficient-conditions-for-a-recursively-defined-sequence-to-be-conv

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
21	Introduction to groups, examples, basic algebra in groups, orders of group elements	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Integrative Approach (**Makes the curriculum more realistic and relatable)
		Mention CLO Number/s	1
		Session Learning Outcome (SLO) [may vary from 1-3]	Understand the concepts of groups, semi-groups, monoids, Congruence relation and quotient structures, Free and cyclic monoids and groups

		URL	https://www.javatpoint.com/discrete-mathematics-group https://www.geeksforgeeks.org/groups-discrete-mathematics/?ref=lbp

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
22	Subgroup, cyclic groups, characteristics of cyclic groups	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Integrative Approach (**Integrates the classroom with the outside world)
		Mention CLO Number/s (may be more than one)	1 and 4
		Session Learning Outcome (SLO) [may vary from 1-3]	Understand the various terminologies associated with graph applications and advanced types
		URL	https://www.geeksforgeeks.org/groups-discrete-mathematics/?ref=lbp https://www.javatpoint.com/discrete-mathematics-subgroup https://www.javatpoint.com/discrete-mathematics-normal-subgroup

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
23	Introduction to Rings and Fields	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Inquiry-Based Approach (**Excites curiosity in the learners)
		Mention CLO Number/s (may be more than one)	1 and 4
		Session Learning Outcome	Understand the need and importance of rings and field structures

		(SLO) [may vary from 1 to 3]	
		URL	https://www.javatpoint.com/lagrange-theorem-in-discrete-mathematics https://www.geeksforgeeks.org/mathematics-rings-integral-domains-and-fields/ https://www.zweigmedia.com/pdfs/146Notes.pdf

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
24	Introduction to recurrence relations and applications of relations	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Constructive Approach (**Learning by doing)
		Mention CLO Number/s	1 and 4
		Session Learning Outcome (SLO) [may vary from 1-3]	Understand the concept of Recurrence Relations and their applicability
		URL	https://www.javatpoint.com/recurrence-relations

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
25	Solving linear recurrent relations	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw Hill
		Pedagogy	**Inquiry-Based Approach (**Enhances problem-solving skills)
		Mention CLO Number/s (may be more than one)	1 and 4

		Session Learning Outcome (SLO)	Understand the concept linear recurrence relations with constant coefficients and solve problems
		URL	https://math.libretexts.org/Bookshelves/Combinatorics_and_Discrete_Mathematics/Discrete_Mathematics_(Levin)/2%3A_Sequences/2.4%3A_Solving_Recurrence_Relations

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
26	Graph and graph models, graph terminology, types of graphs, representations of graph	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw Hill
		Pedagogy	**Constructive Approach** (Builds on past knowledge)
		Mention CLO Number/s (may be more than one)	1 and 2
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Understand the basic definition of graph and its applications
		URL	https://www.javatpoint.com/representation-of-graphs https://www.geeksforgeeks.org/graph-types-and-applications/?ref=lbp

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
27	Graph isomorphism, connectivity, Euler, and Hamiltonian paths	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw Hill
		Pedagogy	**Inquiry-Based Approach** (Excites curiosity in the learners)
		Mention CLO Number/s	1 and 3

		(may be more than one)	
		Session Learning Outcome (SLO	Understand importance of graph and its properties, different identification of terms related to graph (application based)
		URL	https://www.geeksforgeeks.org/mathematics-walks-trails-paths-cycles-and-circuits-in-graph/?ref=lbp https://www.javatpoint.com/isomorphic-and-homeomorphic-graphs

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
28	Problem solving on graphs, Planner graph, and graph coloring	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Inquiry-Based Approach (**Enhances problem-solving skills)
		Mention CLO Number/s (may be more than one)	1 and 3
		Session Learning Outcome (SLO) [may vary from 1 to 3]	Understand the type of graphs and apply the coloring concepts to real-world problems
		URL	https://www.javatpoint.com/discrete-mathematics-dijkstras-algorithm

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
29	Introduction to trees, Applications of trees	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Collaborative Approach (**Pools with different abilities of learners)

		Mention CLO Number/s (may be more than one)	1 and 3
		Session Learning Outcome (SLO) [may vary from 1 to 3]	
		URL	https://www.geeksforgeeks.org/dijkstras-shortest-path-algorithm-greedy-algo-7/?ref=lbp https://www.geeksforgeeks.org/graph-coloring-applications/

Session No.	**Topic/Sub Topic**	**Session Details (Whatever is applicable)**	
30	Spanning trees, minimum spanning trees, and their applications	**Text Book**	C. L. Liu, "Elements of Discrete Mathematics", McGraw-Hill
		Reading (Reference)	Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill
		Pedagogy	**Constructive Approach (**Learning by doing, self-learning)
		Mention CLO Number/s (may be more than one)	1
		Session Learning Outcome (SLO)	Apply the concept of a tree to find a spanning tree
		URL	https://www.javatpoint.com/discrete-mathematics-minimum-spanning-tree https://www.geeksforgeeks.org/prims-minimum-spanning-tree-mst-greedy-algo-5/?ref=lbp https://www.geeksforgeeks.org/kruskals-minimum-spanning-tree-algorithm-greedy-algo-2/?ref=lbp

Session No.	Topic/Sub Topic	Session Details (Whatever is applicable)	
30 (Revisit)	Review of the course, Feedback related to the course, Linkages with advanced course/s in succeeding years.	**Text Book**	-NA-
		Reading (Reference)	-NA-
		Pedagogy	**Inquiry-Based Approach (**Enhances problem-solving skills)
		Mention CLO Number/s (may be more than one)	1, 2, 3, 4
		Session Learning Outcome (SLO) [may vary from 1 to 3]	-NA-
		URL	https://lms.nirmauni.ac.in/course/view.php?id=1067

F. **References**: The Following course material is uploaded on the LMS:

https://lms.nirmauni.ac.in/course/view.php?id=1067

G. **Instructions** (if any)

Students are expected to carry out assigned work under the Continuous Evaluation (CE) component and the Tutorial component independently. Copying in any form is not acceptable and will invite strict disciplinary action. Evaluation of the corresponding component will be affected proportionately in such cases. Turnitin software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of the course

H. **Teaching-learning methodology:**

Lectures: The lectures will be conducted through presentations (PPT). Exhaustive use of Blackboard is desirable for discussion of various concepts of MFCS

I. **Active learning techniques**

Flipped Classroom: The flipped classroom is a pedagogical model in which the typical lecture and homework elements of a course are reversed. Short video lectures are viewed by students at home before the class session, while in-class time is devoted to exercises, projects, or discussions.

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- The Muddiest Point is one of the most straightforward Classroom Assessment Techniques to help assess where students are having difficulties. The technique consists of asking students to jot down a quick response to one question: "What was the muddiest point in the session?"
- Minute Papers
  - Ask students to summarize the topic concisely, completely, and creatively within a minute
  - Asking students to form questions from a given topic within a minute of different weightage (1 mark, 2 marks, 3 marks, and so on)
- Think-Pair-Share

  o Think-pair-share (TPS) is a collaborative learning strategy in which students work together to solve a problem or answer a question about an assigned reading. This technique requires students to

  (1) think individually about a topic or answer to a question; and

  (2) Share ideas with classmates.

  - Problems will be shared in a class to be solved in a group of 2 students

J. **Course Material:**

All relevant information covering various aspects of the course MFCS is put on the website at the link:

Students may go through the blog for the latest information and post their queries 24/7. Other students and subject-related faculty members would be actively participating in their contribution to the course blog. In case of any query or doubt, it is appropriate to post the same on the blog and get it answered either through peers or through concerned faculty members.

K. **Course Outcome Attainment:**

The following means will be used to assess the attainment of course learning outcomes:

- Use of formal evaluation components of continuous evaluation, laboratory work, and semester-end examination
- Informal feedback during the course conduct

L. **Academic Integrity Statement:**

Students are expected to carry out assigned work under the Continuous Evaluation (CE) component. Copying in any form is not acceptable and will invite strict disciplinary action. Evaluation of the corresponding component will be affected proportionately in such cases. Academic integrity is expected from students in all components of the course assessment.